

Ejercicio 13 Guía 1 caso 1

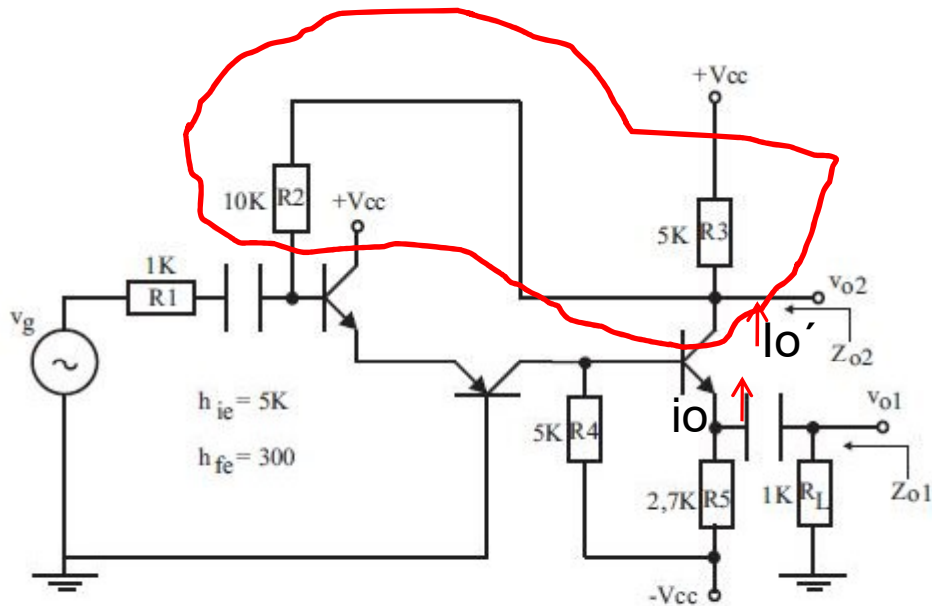


Figura 12: Ejercicio 13.

1- Determinar configuración

Paralelo - Serie

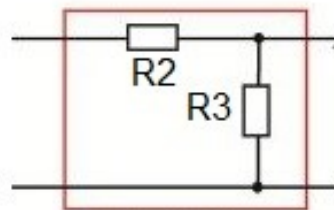
2 - Identificar red de realimentación

Parámetros [g]

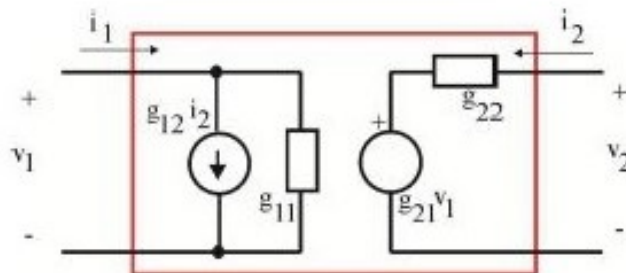
3 – Ganancia requerida y adecuación de entrada de ser necesario

$$i_n = \frac{v_g}{1K} \quad A_i = \frac{i_o}{i_n}$$

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Red de realimentación



Red de realimentación

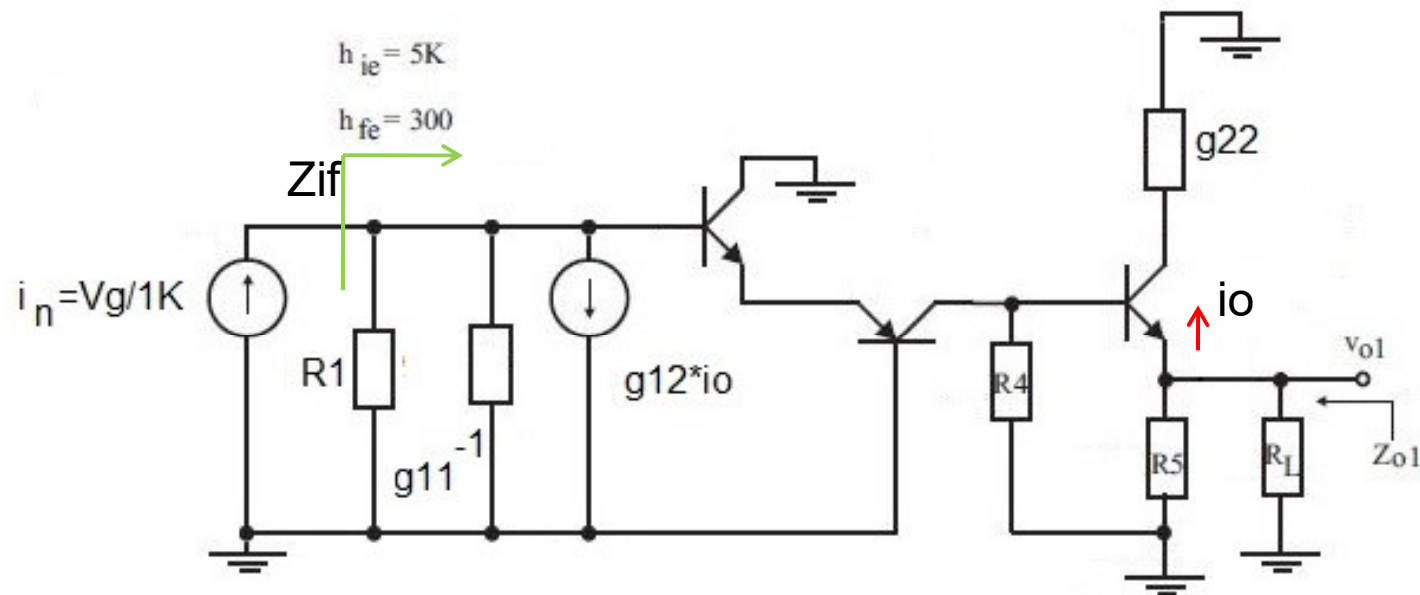
$$g_{11} = \left. \frac{i_1}{v_1} \right|_{i_2=0} = \frac{1}{R2 + R3}$$

$$g_{12} = \left. \frac{i_1}{i_2} \right|_{v_1=0} = -\frac{R3}{R2 + R3}$$

$$g_{21} = \left. \frac{v_2}{v_1} \right|_{i_2=0} = \frac{R3}{R2 + R3}$$

$$g_{22} = \left. \frac{v_2}{i_2} \right|_{v_1=0} = \frac{R2 R3}{R2 + R3}$$

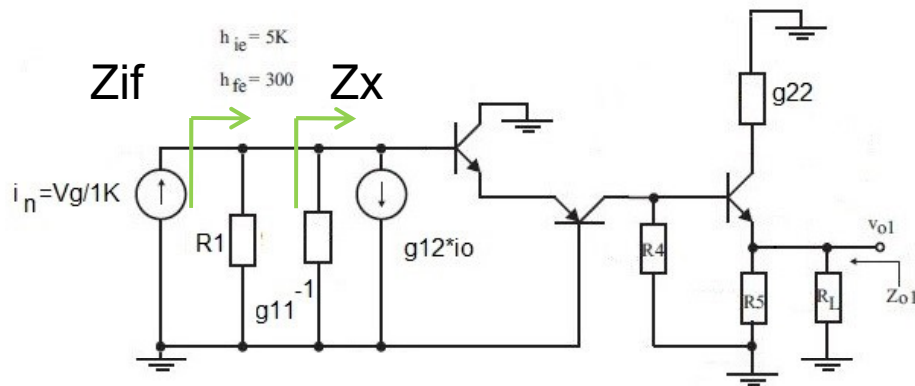
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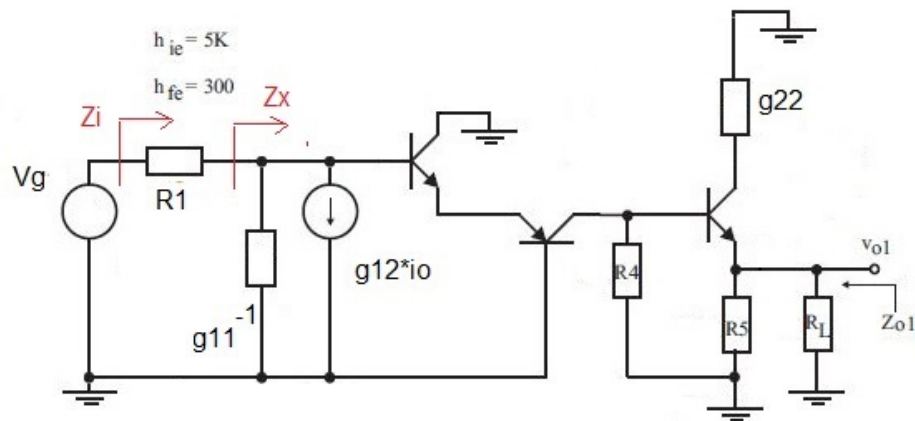
$$a_i = \frac{i_o}{i_n}, \quad \text{con } g_{12} = 0 \quad A_i = \frac{i_o}{i_n} = \frac{a_i}{1 + a_i g_{12}} \quad Z_{if} = \frac{Z_{io}}{1 + a_i g_{12}}$$

$$Z_{io} = R_1 \parallel \left| g_{11}^{-1} \right| (2h_{ie}) \quad A_v = \frac{v_{o1}}{v_g} = \frac{-i_o R_5 \parallel R_L}{i_n 1K} = -A_i \frac{R_5 \parallel R_L}{1K}$$

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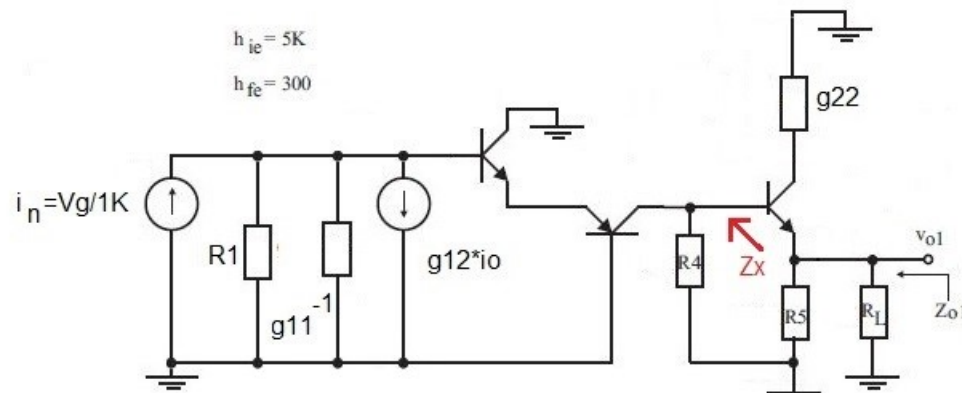


$$Z_{if} = R_1 || Z_x$$

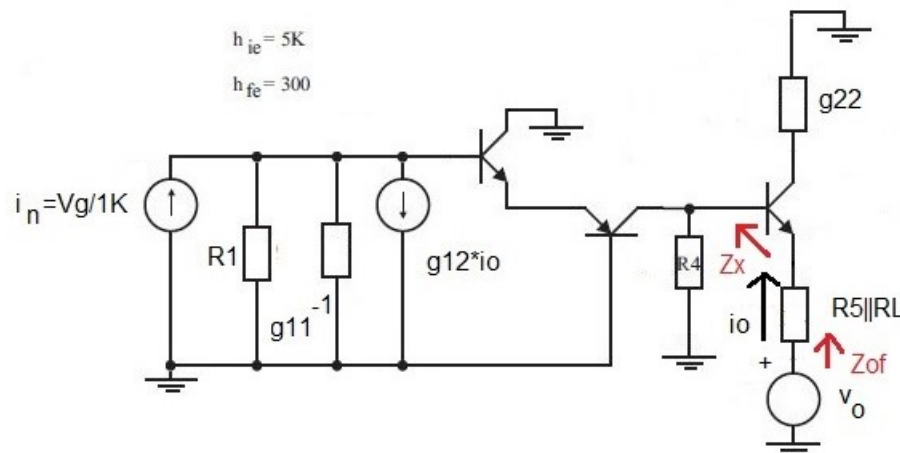


$$Z_i = R_1 + Z_x$$

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$$Z_{o1} = R_5 || R_L || Z_x$$



$$Z_{o0} = R_5 || R_L + \frac{h_{ie} + R_4}{h_{fe} + 1}$$

$$Z_{of} = Z_{o0} (1 + a_i g_{12})$$

$$Z_{of} = R_5 || R_L + Z_x$$